Posture estimation

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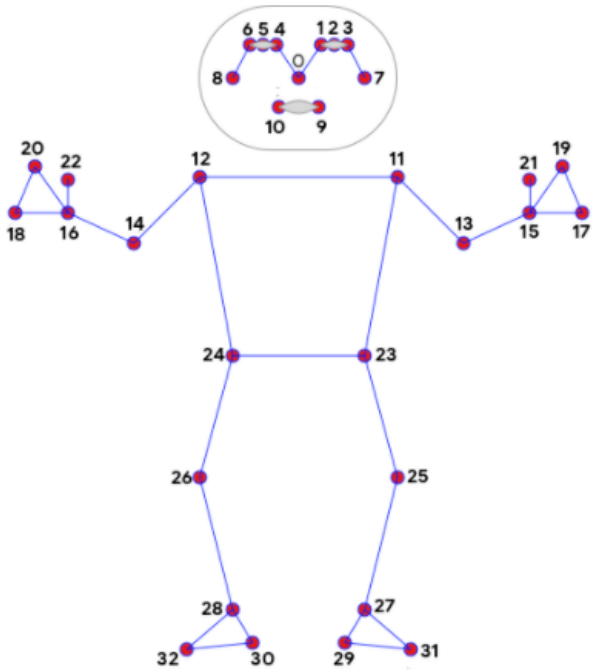
- 1. Posture estimation
- 2. Effect preview
  - 2.1. USB camera

MediaPipe is a cross-platform machine learning framework that can process key point detection such as hands, faces, postures, etc. in videos and images in real time.

1. Posture estimation

MediaPipe Posture estimation can detect and track key points of the human body (such as shoulders, elbows, knees, etc.) and draw the human skeleton.



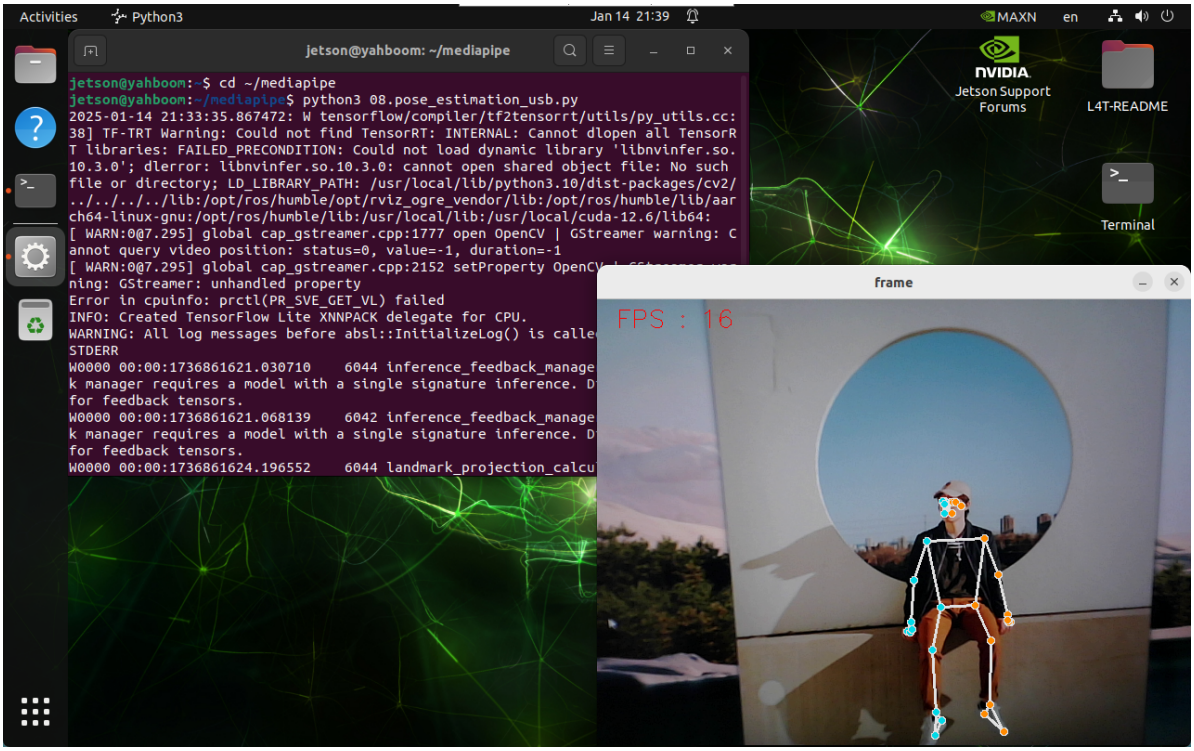
0. nose
1. left_eye_inner
2. left_eye
3. left_eye_outer
4. right_eye_inner
5. right_eye
6. right_eye_outer
7. left_ear
8. right_ear
9. mouth_left
10. mouth_right
11. left_shoulder
12. right_shoulder
13. left_elbow
14. right_elbow
15. left_wrist
16. right_wrist
17. left_pinky
18. right_pinky
19. left_index
20. right_index
21. left_thumb
22. right_thumb
23. left_hip
24. right_hip
25. left_knee
26. right_knee
27. left_ankle
28. right_ankle
29. left_heel
30. right_heel
31. left_foot_index
32. right_foot_index

2. Effect preview

2.1. USB camera

```
cd ~/mediapipe
python3 08.pose_estimation_usb.py
```

Note: Select the camera screen and press `q` to exit the program!


Activities
Python3
Jan 14 21:39
MAXN
en
jetson@yahboom: ~/mediapipe
jetson@yahboom:~$ cd ~/mediapipe
jetson@yahboom:~/mediapipe$ python3 08.pose_estimation_usb.py
2025-01-14 21:33:35.867472: W tensorflow/compiler/tf2tensorrt/utils/py_utils.cc:38] TF-TRT Warning: Could not find TensorRT: INTERNAL: Cannot dlopen all TensorRT libraries: FAILED_PRECONDITION: Could not load dynamic library 'libnvinfer.so.10.3.0'; dlerror: libnvinfer.so.10.3.0: cannot open shared object file: No such file or directory; LD_LIBRARY_PATH: /usr/local/lib/python3.10/dist-packages/cv2/../../../../lib:/opt/ros/humble/opt/rviz_ogre_vendor/lib:/opt/ros/humble/lib/aarch64-linux-gnu:/opt/ros/humble/lib:/usr/local/lib:/usr/local/cuda-12.6/lib64:
[WARN:0@7.295] global cap_gstreamer.cpp:1777 open OpenCV | GStreamer warning: Cannot query video position: status=0, value=-1, duration=-1
[WARN:0@7.295] global cap_gstreamer.cpp:2152 setProperty OpenCV
ning: GStreamer: unhandled property
Error in cpuinfo: prctl(PR_SVE_GET_VL) failed
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
WARNING: All log messages before absl::InitializeLog() is calle
STDERR
W0000 00:00:1736861621.030710 6044 inference_feedback_manage
k manager requires a model with a single signature inference. D
for feedback tensors.
W0000 00:00:1736861621.068139 6042 inference_feedback_manage
k manager requires a model with a single signature inference. D
for feedback tensors.
W0000 00:00:1736861624.196552 6044 landmark_projection_calcu
NVIDIA
Jetson Support Forums
L4T-README
Terminal
frame
FPS : 16